

Instructor: *Avery, Korytov*

PHYSICS DEPARTMENT

PHY2049, Fall 2020

Exam 3

December 2020

Name (print, last first): _____ Signature: _____

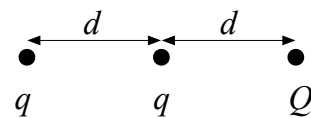
*On my honor, I have neither given nor received unauthorized aid on this examination.***YOUR EXAM NUMBER IS THE 5-DIGIT NUMBER AT THE TOP OF EACH PAGE.****DIRECTIONS**

- (1) **Code your exam number on your answer sheet (use 76–80 for the 5-digit number).** Code your name on your answer sheet. **DARKEN CIRCLES COMPLETELY.** Code your student number on your answer sheet.
- (2) Print your name on this sheet and sign it also.
- (3) Do all scratch work anywhere on this exam that you like. At the end of the exam, this exam printout is to be turned in. No credit will be given without both answer sheet and printout with scratch work most questions demand.
- (4) **Blacken the circle of your intended answer completely, using a #2 pencil or blue or black ink.** Do not make any stray marks or the answer sheet may not read properly.
- (5) The answers are rounded off. Choose the closest to exact. There is no penalty for guessing.

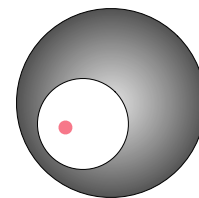
>>>>>>>**WHEN YOU FINISH**<<<<<<<
Hand in the exam and answer sheet separately.

Constants		
$m_e = 9.11 \times 10^{-31} \text{ kg}$	$m_p \simeq m_n = 1.67 \times 10^{-27} \text{ kg}$	$e = 1.6 \times 10^{-19} \text{ C}$
$k = 9 \times 10^9 \text{ N m}^2/\text{C}^2$	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$	$\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$
$\mu_B = 9.27 \times 10^{-24} \text{ J/T}$	$c = 3 \times 10^8 \text{ m/s}$	$g = 9.8 \text{ m/s}^2$
f = femto = 10^{-15}	p = pico = 10^{-12}	n = nano = 10^{-9}
μ = micro = 10^{-6}	M = Mega = 10^6	G = Giga = 10^9
Sphere: $A = 4\pi R^2$	$V = \frac{4\pi}{3} R^3$	

1. Three particles with charges q , q , Q lie on the x -axis a distance d apart as shown in the figure. For what value of Q is the total potential energy of the system equal to $3kq^2/2d$?



- (1) $Q = +\frac{1}{3}q$ (2) $Q = +\frac{1}{2}q$ (3) $Q = +\frac{2}{3}q$ (4) $Q = +q$ (5) $Q = +\frac{1}{4}q$
2. A small charged ball is held in place inside a spherical cavity inside a spherical conductor, as illustrated in the figure. The charge on the inner surface of the conductor is $-5\ \mu\text{C}$ and the charge on the outer surface is $+12\ \mu\text{C}$. What is the charge on the small ball?

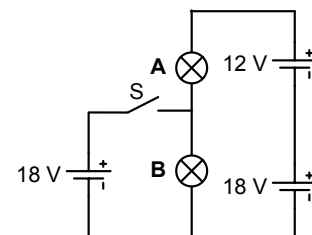


- (1) $+5\ \mu\text{C}$ (2) $+1\ \mu\text{C}$ (3) $-12\ \mu\text{C}$ (4) $-7\ \mu\text{C}$ (5) $+7\ \mu\text{C}$
3. Two isolated conducting spheres are separated by a large distance. Sphere 1 has a radius of 20 cm and an initial charge $-30\ \text{nC}$ while Sphere 2 has a radius of 60 cm and an initial charge $90\ \text{nC}$. A very thin copper wire is now connected to the spheres to allow charge to flow between them. What is the final charge on Sphere 2?

- (1) 45 nC (2) 15 nC (3) 30 nC (4) 60 nC (5) 120 nC
4. What is the magnitude of the electric field at the point $(1, -3, -1)$ if the electric potential in the region is given by $V = x^2z^2$? (V is in volts and all coordinates are in meters.)
- (1) 2.83 V/m (2) 1.41 V/m (3) 2.00 V/m (4) 1.00 V/m (5) 2.23 V/m

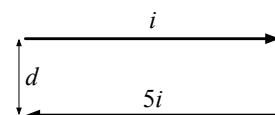
5. A resistor dissipates 160 W when connected to a US power source (110 V, 60 Hz). What is the power dissipated if the resistor is connected to a UK power source (220 V, 50 Hz)?

- (1) 640 W (2) 160 W (3) 40 W (4) 530 W (5) 133 W
6. The light bulbs A and B in the circuit shown are identical. Assume that the switch S has been closed for awhile. What happens to the brightness of the two bulbs after the switch is opened?



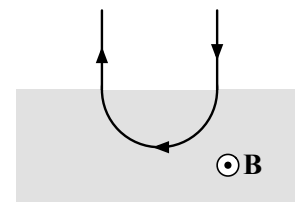
- (1) A gets brighter, B dimmer
(2) A gets dimmer, B brighter
(3) A gets dimmer, B dimmer
(4) A gets brighter, B brighter
(5) Nothing changes

7. Two long parallel wires separated by $d = 12\ \text{cm}$ are shown with the currents indicated. Where does the net B field from both wires equal zero?



- (1) 3 cm above upper wire
(2) 4 cm above upper wire
(3) 4 cm below upper wire
(4) 8 cm above upper wire
(5) 4 cm above lower wire

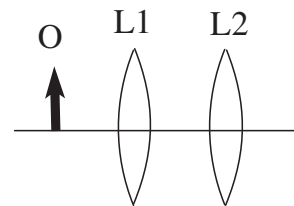
8. In the figure, a charged lithium nucleus (3 protons, 4 neutrons) is accelerated through a potential difference of 10 kV. It enters a region of uniform magnetic field in the direction shown, goes through half a circle of radius 10.0 cm, and then exits the region moving in the opposite direction. What is the radius of the orbit if the same nucleus is accelerated through a potential difference of 30 kV?



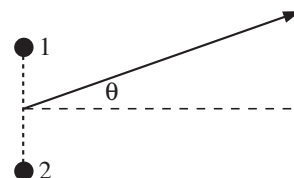
- (1) 17.3 cm (2) 3.3 cm (3) 10.0 cm (4) 30 cm (5) 8.66 cm
9. A $50\ \mu\text{F}$ capacitor with an initial potential difference of 100 V is discharged through a resistor after a switch between them is closed. If the current in the circuit is 10 mA immediately after the switch is closed, what is the voltage across the capacitor 0.8 s later?
- (1) 20 V (2) 30 V (3) 37 V (4) 45 V (5) 100 V
10. A conductive loop is placed next to a wire carrying current I as shown. If the current is *increasing*, what is the direction of the net force on the loop?
- (1) right
(2) left
(3) up
(4) down
(5) net force = 0
11. A hemispherical volume of diameter 80 cm is placed in a constant B field of magnitude 3.0 T pointing vertically upward as shown. The flat face normal is 40° from vertical. What is the magnetic flux through the curved surface?
- (1) $-1.16\ \text{T}\cdot\text{m}^2$
(2) $-0.97\ \text{T}\cdot\text{m}^2$
(3) $+2.31\ \text{T}\cdot\text{m}^2$
(4) $-1.94\ \text{T}\cdot\text{m}^2$
(5) $+0.97\ \text{T}\cdot\text{m}^2$
12. An isolated parallel plate capacitor with two plates of radius 40 cm and plate separation $1.0\ \mu\text{m}$ has an initial voltage of 120 V between the plates. The capacitor is then connected to a $100\ \Omega$ resistor. Immediately after the connection, what is the magnitude of the B field inside the capacitor at a radius of 30 cm from the center?
- (1) $0.45\ \mu\text{T}$ (2) $0.60\ \mu\text{T}$ (3) $0.30\ \mu\text{T}$ (4) $1.20\ \mu\text{T}$ (5) $5.5\ \mu\text{T}$
13. A laser beam of power 4.60 W and diameter 2.60 mm is directed upward and shines perpendicularly onto a flat face of a fully-reflecting cube of side L and density $2700\ \text{kg}/\text{m}^3$. What is the largest value of L for which the force due to the laser light pressure can overcome the Earth's gravitational force? (Assume that the cube fits inside the laser beam.)
- (1) $0.22\ \mu\text{m}$ (2) $0.60\ \mu\text{m}$ (3) $0.055\ \mu\text{m}$ (4) $2.1\ \mu\text{m}$ (5) $0.11\ \mu\text{m}$
14. Refer to the previous problem. If the laser beam has a wavelength of 630 nm, what is the maximum value of the B field propagated in the beam?
- (1) $85\ \mu\text{T}$ (2) $60\ \mu\text{T}$ (3) $43\ \mu\text{T}$ (4) $25\ \mu\text{T}$ (5) $55\ \mu\text{T}$
15. A makeup mirror has radius of curvature 54 cm. How far is the upright image from your face if it is 6 times larger?
- (1) 158 cm (2) 45 cm (3) 315 cm (4) 22.5 cm (5) 135 cm

16. In the figure shown (not to scale), the focal lengths of the thin converging lenses, L1 and L2, are 20 cm and 10 cm, respectively. When object O is placed 30 cm to the left of L1, its image forms at 30 cm to the right of L2. What is the distance between the two lenses?

(1) 75 cm
 (2) 30 cm
 (3) 20 cm
 (4) 50 cm
 (5) 40 cm



17. As shown in the figure, two radio towers are spaced a distance of 21 m apart and radiating at a wavelength of 4.5 m. Tower 2 radiates at the same intensity as tower 1 but 48° later in phase. If a radio receiver is placed a long distance away, what is the smallest angle θ at which the intensity of the combined signal is a relative minimum?



(1) 4.5° (2) 6.2° (3) 10.7° (4) 12.4° (5) 29.6°

18. A thin film of thickness 280 nm and index of refraction 1.4 coats a plate of glass having index of refraction 1.65. If light is perpendicularly incident on the glass, at what wavelength is the reflected light minimal?

(1) 523 nm (2) 392 nm (3) 261 nm (4) 784 nm (5) 373 nm

19. A consortium plans to build a 30-meter diameter telescope having unprecedented light collection capability and angular resolution. If the angular resolution is limited by diffraction, what is the approximate maximum distance in light years ($1 \text{ light year} = 9.5 \times 10^{12} \text{ km}$) that the telescope can distinguish two stars separated by the earth-sun distance of $1.5 \times 10^8 \text{ km}$? Assume that the telescope uses 1100 nm infrared light.

(1) 350 (2) 1050 (3) 700 (4) 850 (5) 120

20. A slit 1.00 mm wide is illuminated by light of wavelength 450 nm. We see a diffraction pattern on a screen 2.00 m away. What is the distance between the first two diffraction minima on the same side of the central diffraction maximum?

(1) 0.90 mm (2) 0.45 mm (3) 1.35 mm (4) 1.80 mm (5) 2.25 mm